How can we group diagrams? including non-BCJ

Once we set a K-zero, we split the X_{ij} in constrained and born-free
boundary propagators

①. group by # of boundary propagators (group by B_i terms (that scale like z^i))

②. group by choice of born free → D subsets

- ② includes ①

group by means that each group must be zero independently.

y : constrained variables

z : free variables

$$\frac{1}{y^\alpha z^\beta}$$

$B = B(y,z)$

aside

impose constraints on y $\left(L(y)=0 \right)$

$y \to \lambda y$ still satisfy constraint

so $0 = B(\lambda y, z)$

For ② if X_{35} X_{36} are born free, then all the terms that have $\frac{1}{X_{35}X_{36}}$ and no other born free can be factored together.

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easy consequences:

- terms that have only born free variables are killed

D subset has 1 element

- 1 boundary propagator ($n-4$ born free)

→ factor out the born free → left with linear combination

$$\sum_{\substack{(i,j)\in \text{boundary prop.} \\ i<j}} \frac{a_{ij}}{X_{ij}} = 0$$

ex : 1-zero

$$\frac{e_{13}}{X_{13}}+\frac{e_{14}}{X_{14}}+\dots+\frac{e_{1n-1}}{X_{1n-1}}+\frac{e_{24}}{X_{24}}+\frac{e_{25}}{X_{25}}+\dots+\frac{e_{2n}}{X_{2n}}=0$$

$X_{13} \to \infty$

$X_{2n} = -X_{13} \to -\infty$

send everything but X_{14} to ∞ $\to$ $e_{14}=0$

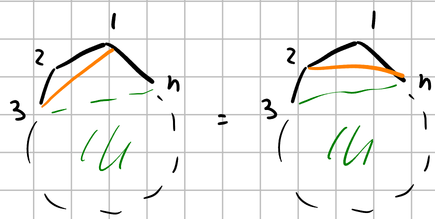
send everything but X_{24} to ∞ $\to$ $e_{24}=0$

$X_{24} = X_{14} - X_{13}$

etc $\to$ all that's left is $\frac{e_{13}}{X_{13}}+\frac{e_{2n}}{X_{2n}}=0$

since $X_{2n}=-X_{13}$ we get $\frac{1}{X_{13}}(e_{13}-e_{2n})=0$ so $\boxed{e_{13}=e_{2n}}$

what about 2 boundary props ?



$$\sum_{(i_1 j_1),(i_2 j_2)\in BP} \frac{e_{i_1j_1\, i_2j_2}}{X_{i_1j_1}\,X_{i_2j_2}}=0$$

$i_1 \le i_2$, $j_1 \le j_2$ $\to$ avoid duplicates

$$\frac{e_{1313}}{X_{13}^2}+\frac{e_{1314}}{X_{13}X_{14}}+\dots+\frac{e_{13\,1n-1}}{X_{13}X_{1n-1}}+\frac{e_{1324}}{X_{13}X_{24}}+\frac{e_{1325}}{X_{13}X_{25}}+\dots+\frac{e_{13\,2n}}{X_{13}X_{2n}}+$$

$$\frac{e_{1414}}{X_{14}^2}+\dots+\frac{e_{14\,1n-1}}{X_{14}X_{1n-1}}+\frac{e_{1424}}{X_{14}X_{24}}+\frac{e_{1425}}{X_{14}X_{25}}+\dots+\frac{e_{14\,2n}}{X_{14}X_{2n}}+$$

$$\frac{e_{1515}}{X_{15}^2}+\dots+\frac{e_{15\,1n-1}}{X_{15}X_{1n-1}}+\frac{e_{1524}}{X_{15}X_{24}}+\frac{e_{1325}}{X_{15}X_{25}}+\dots+\frac{e_{15\,2n}}{X_{15}X_{2n}}+$$

$$\dots+\frac{e_{2n\,2n}}{X_{2n}^2}$$

send $X_{13}, X_{2n} \to \infty$

can choose one from each (X_{1i}, X_{2i}) $i=4,\dots,n-1$ to go to ∞

$e_{1414}=0$ $e_{14\,15}=0$ $e_{14\,16}=0$ etc.

things that survive: $a_{1j\,2j}$, $a_{13\,1j}$, $a_{13\,2j}$, $a_{2n\,1j}$, $a_{2n\,2j}$ $\qquad j=4,\dots,n-1$

$a_{13\,13}$, $a_{2n\,2n}$, $a_{13\,2n}$

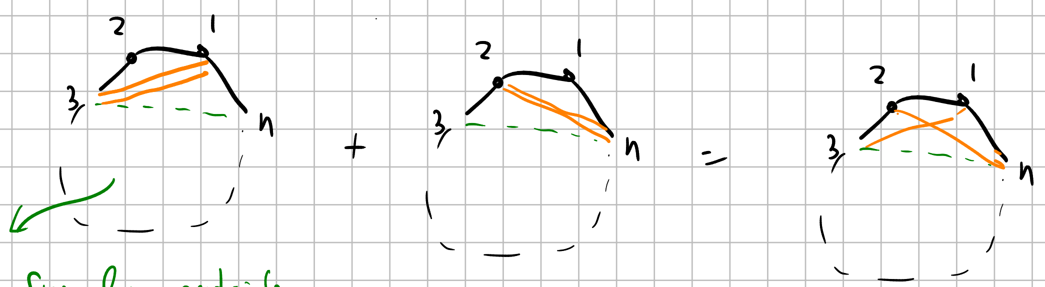
send everything but X_{13}, X_{2n} to ∞ $\qquad \underbrace{X_{2j}}_{\to\infty} = \underbrace{X_{1j}}_{\to\infty} - X_{13}$

$$\Rightarrow \frac{\underbrace{a_{13\,13}+a_{2n\,2n}-a_{13\,2n}}}{X_{13}^2} = 0 \qquad (X_{2n}=-X_{13})$$

$$\boxed{a_{13\,2n} = a_{13\,13}+a_{2n\,2n}}$$

all crossings

Note: abuse of notation since each $a_{ij\,kl}$ depends on choice of born free. if α is a multi index rep. born free ($n-5$ of them) then we really should write $a_{\alpha\, i_1 j_1\, i_2 j_2}$



born free line outside of the 4-gon

(born free in all pictures are the same)

try sending stuff to zero $\quad (*)$

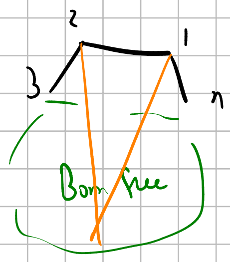
$$\frac{a_{13\,13}}{X_{13}^2}+\frac{a_{2n\,2n}}{X_{2n}^2}+\sum_{j=4}^{n-1}\frac{1}{X_{13}}\left(\frac{a_{13\,1j}}{X_{1j}}+\frac{a_{13\,2j}}{X_{2j}}\right)+\sum_{j=4}^{n-1}\frac{1}{X_{2n}}\left(\frac{a_{2n\,1j}}{X_{1j}}+\frac{a_{2n\,2j}}{X_{2j}}\right)+\sum_{j=4}^{n-1}\frac{a_{1j\,2j}}{X_{1j}X_{2j}}$$
$$+\frac{a_{13\,2n}}{X_{13}X_{2n}}$$

common denominator: $\quad X_{13}^2X_{2n}^2\left(\prod_{j=4}^{n-1}X_{1j}X_{2j}\right)$

numerator:

$$a_{13\,13}\,X_{2n}^2\prod_j X_{1j}X_{2j} + a_{2n\,2n}\,X_{13}^2\prod_j X_{1j}X_{2j} + a_{13\,2n}\,X_{13}X_{2n}\prod_j X_{1j}X_{2j}$$
$$+\sum_\ell a_{13\,1\ell}\,X_{13}X_{2n}^2\left(\prod_{j\neq\ell}X_{1j}\right)\prod_j X_{2j} + \sum_\ell a_{13\,2\ell}\,X_{13}X_{2n}^2\prod_j X_{1j}\prod_{j\neq\ell}X_{2j}$$
$$+\sum_\ell a_{2n\,1\ell}\,X_{2n}X_{13}^2\left(\prod_{j\neq\ell}X_{1j}\right)\prod_j X_{2j} + \sum_\ell a_{2n\,2\ell}\,X_{2n}X_{13}^2\prod_j X_{1j}\prod_{j\neq\ell}X_{2j}$$
$$+\sum_\ell a_{1\ell\,2\ell}\,X_{13}^2X_{2n}^2\left(\prod_{j\neq\ell}X_{1j}\right)\left(\prod_{j\neq\ell}X_{2j}\right) = 0$$

send $X_{1\ell} \to 0$, $X_{2\ell} \to 0$ then X_{13} must also be zero! $(X_{2\ell} = X_{1\ell} - X_{13})$



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send only either $X_{1\ell}$ or $X_{2\ell}$ to zero

if $X_{1\ell} \to 0$

$$a_{13\,1\ell} \; X_{13} X_{2n}^2 \left(\prod_{j\neq \ell} X_{1j}\right) \prod_{j} X_{2j} \; +$$

$$a_{2n\,1\ell} \; X_{2n} X_{13}^2 \left(\prod_{j\neq \ell} X_{1j}\right) \prod_{j} X_{2j} \; +$$

$$a_{1\ell\,2\ell} \; X_{13}^2 X_{2n}^2 \left(\prod_{j\neq \ell} X_{1j}\right) \left(\prod_{j\neq \ell} X_{2j}\right) = 0$$

$$\to \quad 0 = a_{13\,1\ell} \; X_{2n} X_{2\ell} + a_{2n\,1\ell} \; X_{13} X_{2\ell} + a_{1\ell\,2\ell} \; X_{13} X_{2n}$$

still have $X_{2n} = -X_{13}$, $X_{2\ell} = -X_{13}$

send $X_{13} \to 1$, $X_{2n} \to -1$, $X_{2\ell} \to -1$

$$\boxed{a_{13\,1\ell} = a_{2n\,1\ell} + a_{2\ell\,1\ell}}$$

if $X_{2\ell} \to 0$

$$a_{13\,2\ell} \; X_{13} X_{2n}^2 \prod_{j} X_{1j} \prod_{j\neq \ell} X_{2j}$$

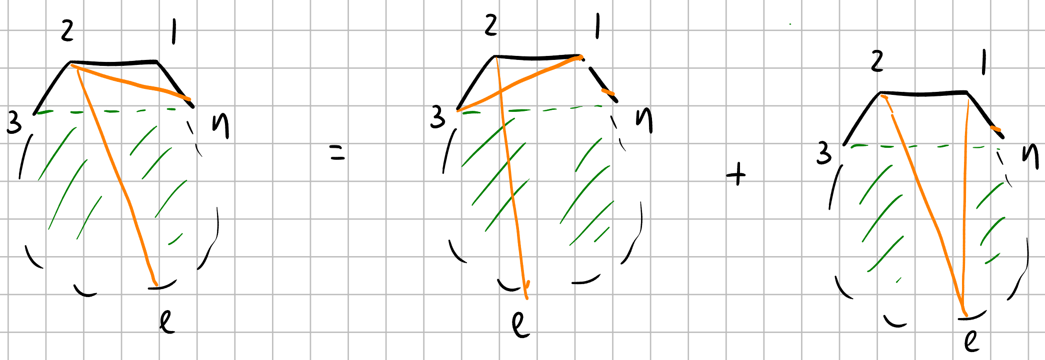
$$+ \; a_{2n\,2\ell} \; X_{2n} X_{13}^2 \prod_{j} X_{1j} \prod_{j\neq \ell} X_{2j}$$

$$+ \; a_{1\ell\,2\ell} \; X_{13}^2 X_{2n}^2 \left(\prod_{j\neq \ell} X_{1j}\right) \left(\prod_{j\neq \ell} X_{2j}\right) = 0$$

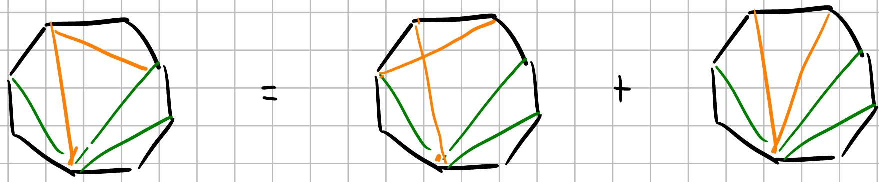
$$0 = a_{13\,2\ell} \; X_{1\ell} X_{2n} + a_{2n\,2\ell} \; X_{1\ell} X_{13} + a_{1\ell\,2\ell} \; X_{13} X_{2n}$$

$X_{13} \to 1$ $\quad X_{1\ell} \to 1$ $\quad X_{2n} \to -1$ $\quad (X_{2\ell} = X_{1\ell} - X_{13})$

$$\boxed{a_{2n\,2\ell} = a_{13\,2\ell} + a_{1\ell\,2\ell}}$$



ex:



consistent

3 boundary propagators

$$\sum_{|\beta|=3} \frac{a_{\alpha\beta}}{y^{\beta}} = 0$$

multi-indices:

β multi-index, y is a vector containing all boundary propagators

α multi index with $|\alpha| = n-6$ (multiplicities of bdry free)

which "a" survives? we must have one of these options:

- X_{13} (one or more)
- X_{2n}
- both X_{1l} and X_{2l}

$\boxed{a_{131313} \quad a_{1313\,2n} \quad a_{13\,2n\,2n} \quad a_{2n\,2n\,2n}}$

$a_{1313\,1l}$ $a_{1313\,2l}$ $a_{13\,2n\,1l}$ $a_{13\,2n\,2l}$ $a_{13\,1l_1\,1l_2}$ $a_{13\,1l_1\,2l_2}$ $a_{13\,2l_1\,2l_2}$

$a_{2n\,2n\,1l}$ $a_{2n\,2n\,2l}$ $a_{2n\,1l\,1j}$ $a_{2n\,1l\,2j}$ $a_{2n\,2l\,2j}$

$a_{1l\,2l\,1j}$ $a_{1l\,2l\,2j}$

$l, j \in \{4, \dots, n-1\}$

send everything but X_{13}, X_{2n} to ∞

$\rightarrow$ $a_{131313} - a_{1313\,2n} + a_{13\,2n\,2n} - a_{2n\,2n\,2n} = 0$

Common denominator: $X_{13}^3 X_{2n}^3 \left(\prod_i X_{1i}^2 X_{2i}^2\right)$

sending 13, 2n to zero doesn't work like before

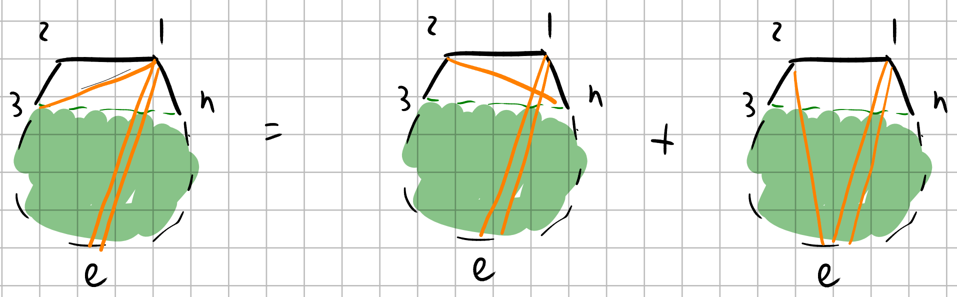
send $X_{1l} \to 0$: the only terms that survive are the ones that contain

X_{1l} twice, so $a_{13\,1l\,1l}$ $a_{2n\,1l\,1l}$ $a_{1l\,2l\,1l}$

$$a_{13\,1l\,1l}\, X_{2n} X_{2l} + a_{2n\,1l\,1l}\, X_{13} X_{2l} + a_{1l\,2l\,1l}\, X_{13} X_{2n} = 0$$

$X_{13} \to 1$ $\quad X_{2n} \to -1$ $\quad X_{2l} \to -1$

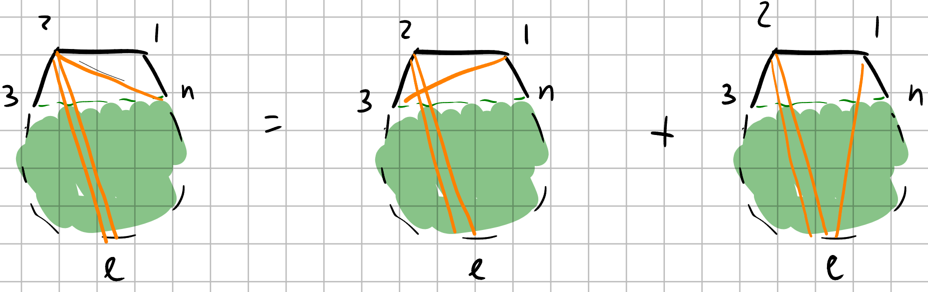
$$\boxed{a_{13\,1l\,1l} = a_{2n\,1l\,1l} + a_{2l\,1l\,1l}}$$



$X_{2l} \to 0$

$$a_{13\,2l\,2l}\, X_{2n} X_{1l} + a_{2n\,2l\,2l}\, X_{13} X_{1l} + a_{1l\,2l\,2l}\, X_{13} X_{2n} = 0$$

$$\boxed{a_{2n\,2l\,2l} = a_{13\,2l\,2l} + a_{1l\,2l\,2l}}$$



$\boxed{\text{2-zero ?}}$

1 BP

BP: $X_{14}, X_{15}, \ldots X_{1\,n-1}$

$X_{35}, \ldots, X_{3n}$

$X_{24}, \ldots, X_{2n}$

$$\frac{e_{14}}{X_{14}} + \frac{e_{15}}{X_{15}} + \ldots \frac{e_{1\,n-1}}{X_{1\,n-1}} + \frac{e_{35}}{X_{35}} + \ldots + \frac{e_{3n}}{X_{3n}} + \frac{e_{24}}{X_{24}} + \ldots + \frac{e_{2n}}{X_{2n}}$$

can send everything but X_{1j} $(j=5,\ldots,n-1)$ to ∞ so $e_{1j}=0$

can send everything but X_{3j} $(j=5,\ldots,n-1)$ to ∞ so $e_{3j}=0$

can send everything but X_{2j} $(j=4,\ldots,n)$ to ∞ so $e_{24}=\ldots=e_{2n}=0$

$$\rightarrow \frac{e_{14}}{X_{14}} + \frac{e_{3n}}{X_{3n}} = 0 \rightarrow \boxed{e_{14} = e_{3n}}$$

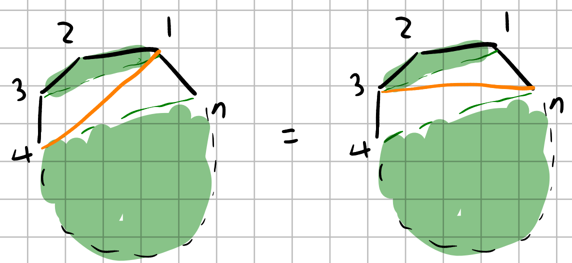
what is free?

$46, 47, \ldots 4n$

$57 \ldots 5n$

$\ldots n-2,n$

13



green regions contain $n-4$ chords

(same on both sides)

for $n=1$ the born free are the chords that don't cross 13 and $2n$

for $k=2$, the born free are the chords that don't cross 14 and $3n$

2 BP

$X_{14}, X_{15}, - X_{1\,n-1}$

$X_{35}, ---, X_{3n}$

$X_{24}, ---, X_{2n}$

if only one of these spheres the term is killed
- 24, 15, --, 1 n-1
- 2n, 35, ---, 3 n-1
- one each of $(1j, 2j, 3j)$ $j = 5, --, n-1$

only terms that survive are

$e_{14\,14}$ $e_{3n\,3n}$ $e_{14\,3n}$

$e_{14\,1j}$ $e_{14\,3j}$ $e_{14\,2j}$ $e_{14\,24}$ $e_{14\,2n}$

$e_{3n\,1j}$ $e_{3n\,3j}$ $e_{14\,2j}$ $e_{3n\,24}$ $e_{3n\,2n}$

$e_{1j\,2j}$ $e_{1j\,3j}$ $e_{2j\,3j}$

$e_{24\,2n}$ $e_{24\,2j}$ $e_{24\,3j}$ $e_{14\,24}$ $e_{3n\,24}$

$e_{2n\,1j}$ $e_{2n\,2j}$ $e_{14\,2n}$ $e_{3n\,2n}$

Can we send everything but X_{14} and X_{3n} to ∞ ?

no, but we can send everything but $X_{14}, X_{3n}, X_{24}, X_{2n}$ to ∞

$X_{35} = X_{15} - X_{14}$

$\vdots$

$X_{3n-1} = X_{1n-1} - X_{14}$

$X_{3n} = - X_{14}$

$X_{25} = X_{35} + X_{24}$

$\vdots$

$X_{2n} = X_{3n} + X_{24}$

$X_{3n} = -X_{14}$

$X_{2n} = -X_{14} + X_{24}$ → arbitrary

→

$$\frac{e_{14\,14}}{X_{14}^2} + \frac{e_{3n\,3n}}{X_{3n}^2} + \frac{e_{14\,3n}}{X_{14}X_{3n}} + \frac{e_{14\,24}}{X_{14}X_{24}} + \frac{e_{3n\,24}}{X_{3n}X_{24}} + \frac{e_{3n\,2n}}{X_{3n}\;X_{2n}} + \frac{e_{14\,2n}}{X_{14}\;X_{2n}} + \frac{e_{24\,2n}}{X_{24\,2n}}$$

Common denominator

$$X_{14}^2\,X_{3n}^2\,X_{24}\,X_{2n}$$

$X_{24} \to 0$ $\qquad e_{14\,24} X_{2n} X_{3n} + e_{3n\,24} X_{2n} X_{14} + e_{2n\,24} X_{14} X_{3n} = 0$

$X_{3n} = -X_{14}$ $\quad \to \quad \boxed{e_{14\,24} = e_{3n\,24} + e_{2n\,24}}$ $\leftarrow$ $K = 2$

$X_{2n} = -X_{14}$

$K = 1$

$\boxed{e_{13\,2n} = e_{13\,13} + e_{2n\,2n}}$

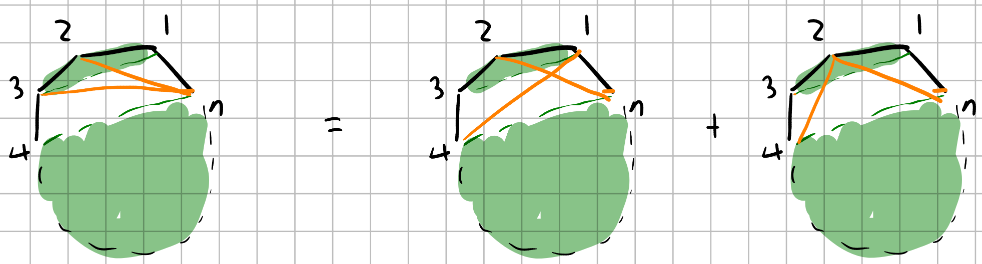
$X_{2n} \to 0$

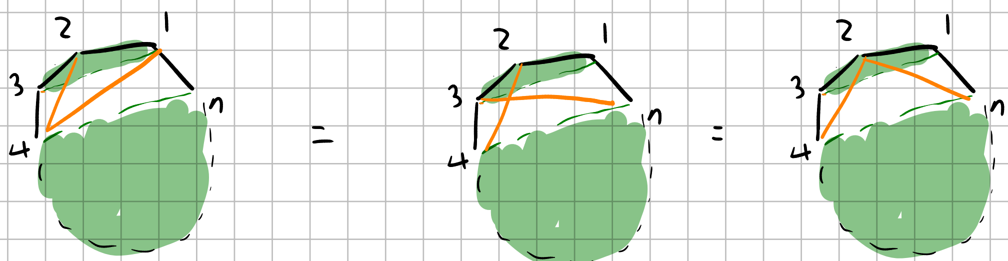
$$e_{14\,2n} X_{24} X_{3n} + e_{3n\,2n} X_{24} X_{14} + e_{2n\,24} X_{14} X_{3n} = 0$$

$X_{3n} = -X_{14}$

$X_{24} = X_{14}$

$\boxed{e_{3n\,2n} = e_{14\,2n} + e_{2n\,24}}$





$X_{3n} = -X_{14}$

$X_{2n} = -X_{14} + X_{24}$ $\to$ arbitrary

$$\frac{e_{14\,14}}{X_{14}^2} + \frac{e_{3n\,3n}}{X_{3n}^2} + \frac{e_{14\,3n}}{X_{14} X_{3n}} + \frac{e_{14\,24}}{X_{14} X_{24}} + \frac{e_{3n\,24}}{X_{3n} X_{24}} + \frac{e_{3n\,2n}}{X_{3n} X_{2n}} + \frac{e_{14\,2n}}{X_{14} X_{2n}} + \frac{e_{24\,2n}}{X_{24}\,X_{2n}}$$

try $\quad X_{14} = 1 \quad X_{3n} = -1 \quad X_{24} = -1 \quad X_{2n} = -2$

$$0 = e_{14\,14} + e_{3n\,3n} - e_{14\,3n} - e_{14\,24} + e_{3n\,24} + \frac{1}{2} e_{3n\,2n} - \frac{1}{2} e_{14\,2n} + \frac{1}{2} e_{24\,2n}$$

plug in $e_{3n\,2n} = e_{14\,2n} + e_{2n\,24}$

$$0 = e_{14\,14} + e_{3n\,3n} - e_{14\,3n} - e_{14\,24} + e_{3n\,24} + \frac{1}{2}\left(e_{14\,2n} + e_{24\,2n} - e_{14\,2n} + e_{24\,2n}\right)$$

$$\boxed{e_{14\,14} + e_{3n\,3n} - e_{14\,3n} - e_{14\,24} + e_{3n\,24} + e_{24\,2n} = 0}$$

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other approach:

$$0 = \frac{e_{14\,14}}{X_{14}^2} + \frac{e_{3n\,3n}}{X_{3n}^2} + \frac{e_{14\,3n}}{X_{14}X_{3n}} + \frac{e_{14\,24}}{X_{14}X_{24}} + \frac{e_{3n\,24}}{X_{3n}X_{24}} + \frac{e_{3n\,2n}}{X_{3n}X_{2n}} + \frac{e_{14\,2n}}{X_{14}X_{2n}} + \frac{e_{24\,2n}}{X_{24\,2n}}$$

when $X_{3n} = -X_{14}$

$X_{2n} = -X_{14} + X_{24}$

if we consider X_{24} arbitrary, then we get a polynomial in X_{24}
all terms must be zero independently.

$X_{14} = \lambda$ $\quad X_{3n} = -\lambda$ $\quad X_{24} = \alpha$ $\quad X_{2n} = \alpha - \lambda$

$$0 = \frac{e_{14\,14}}{\lambda^2} + \frac{e_{3n\,3n}}{\lambda^2} - \frac{e_{14\,3n}}{\lambda^2} + \frac{e_{14\,24}}{\alpha\lambda} - \frac{e_{3n\,24}}{\alpha\lambda} - \frac{e_{3n\,2n}}{\lambda(\alpha-\lambda)} + \frac{e_{14\,2n}}{\lambda(\alpha-\lambda)} + \frac{e_{24\,2n}}{\alpha(\alpha-\lambda)}$$

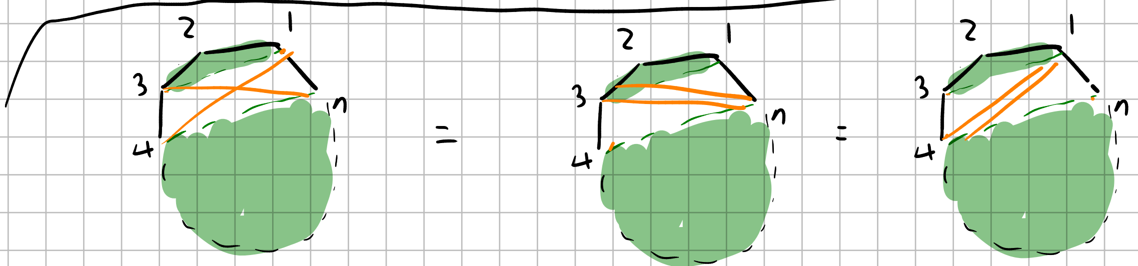
common denominator $\lambda^2\alpha(\alpha-\lambda)$

$$(e_{14\,14} + e_{3n\,3n} - e_{14\,3n})\,\alpha(\alpha-\lambda) + (e_{14\,24} - e_{3n\,24})\,\lambda(\alpha-\lambda)$$
$$+ (e_{14\,2n} - e_{3n\,2n})\,\lambda\alpha + e_{24\,2n}\,\lambda^2$$

$\alpha \to 0$ $\quad e_{14\,24} - e_{3n\,24} - e_{24\,2n} = 0$

$\lambda \to 0$ $\quad \boxed{e_{14\,14} + e_{3n\,3n} - e_{14\,3n} = 0}$ new

$\alpha = \lambda$ $\quad e_{14\,2n} - e_{3n\,2n} + e_{24\,2n} = 0$



Review case ⊛ (2 BP on $k=1$)

$$\frac{e_{1313}}{X_{13}^2}+\frac{e_{2n2n}}{X_{2n}^2}+\sum_{j=4}^{n-1}\frac{1}{X_{13}}\left(\frac{e_{131j}}{X_{1j}}+\frac{e_{132j}}{X_{2j}}\right)+\sum_{j=4}^{n-1}\frac{1}{X_{2n}}\left(\frac{e_{2n1j}}{X_{1j}}+\frac{e_{2n2j}}{X_{2j}}\right)+\sum_{j=4}^{n-1}\frac{e_{1j2j}}{X_{1j}X_{2j}}$$
$$+\frac{e_{132n}}{X_{13}X_{2n}}=0$$

new trick: set $X_{2n}=-X_{13}$ first, with $1j$ and $2j$ still indep.

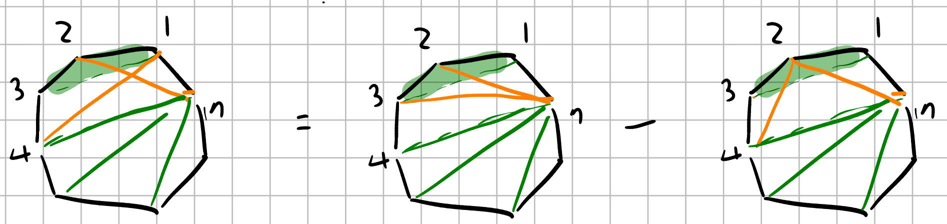
$$\frac{e_{1313}}{X_{13}^2}+\frac{e_{2n2n}}{X_{13}^2}+\sum_{j=4}^{n-1}\frac{1}{X_{13}}\left(\frac{e_{131j}}{X_{1j}}+\frac{e_{132j}}{X_{2j}}\right)-\sum_{j=4}^{n-1}\frac{1}{X_{13}}\left(\frac{e_{2n1j}}{X_{1j}}+\frac{e_{2n2j}}{X_{2j}}\right)$$
$$+\sum_{j=4}^{n-1}\frac{e_{1j2j}}{X_{1j}X_{2j}}-\frac{e_{132n}}{X_{13}^2}=0$$

<u>Common denominator:</u> $X_{13}^2\prod_j X_{1j}X_{2j}$

send $X_{13}\to 0$: $\boxed{e_{1313}+e_{2n2n}-e_{132n}=0}$

send $X_{1j}\to 0$, $X_{2j}\to -X_{13}$ $\quad \frac{e_{131j}}{X_{13}}-\frac{e_{2n1j}}{X_{13}}+\frac{e_{1j2j}}{X_{2j}}=0 \rightarrow \boxed{e_{131j}-e_{2n1j}-e_{1j2j}=0}$

send $X_{2j}\to 0$ (done before)



<u>idea:</u> can we show that all bdls have same coefficient (without first showing that non-bdls vanish)?